



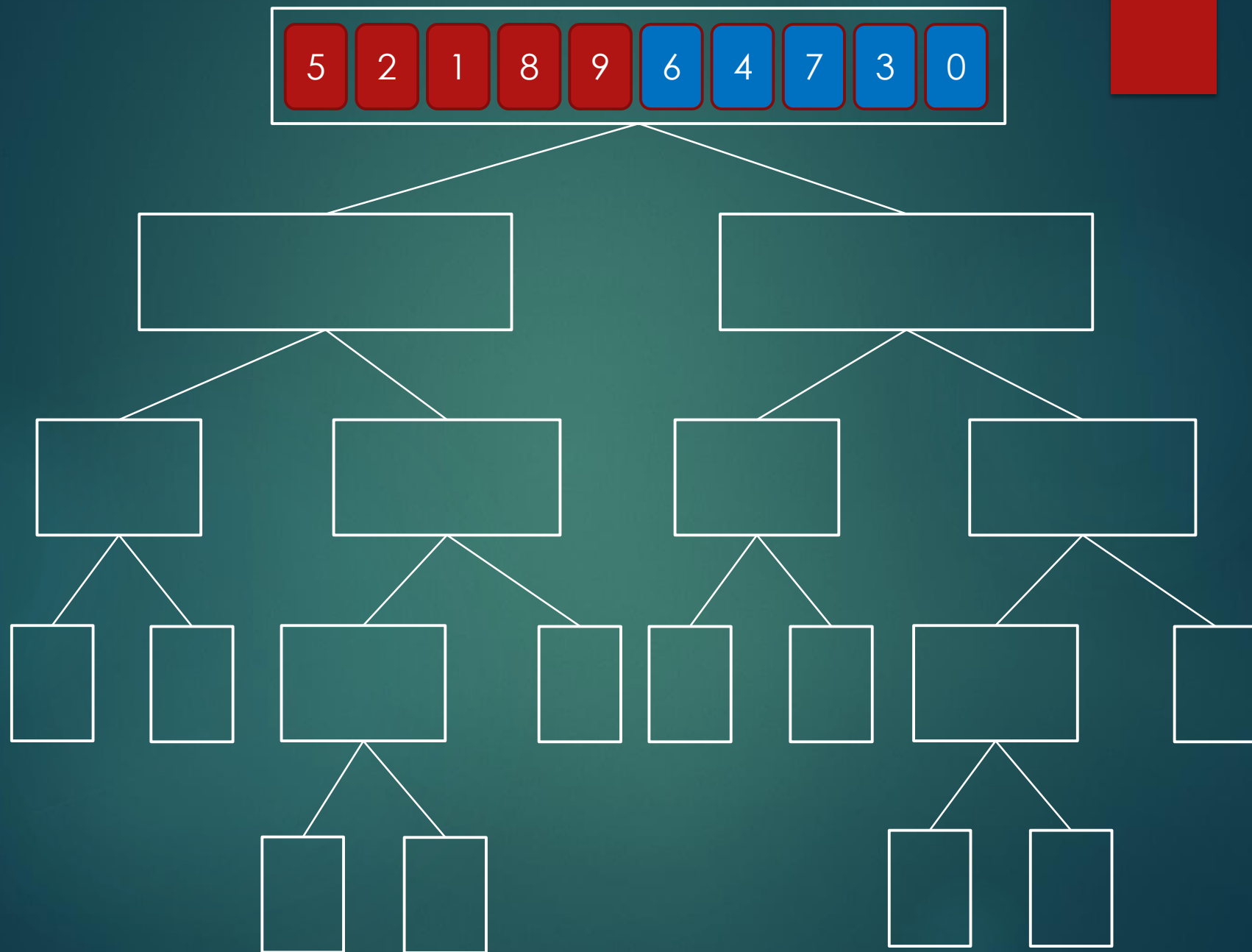
LT12d

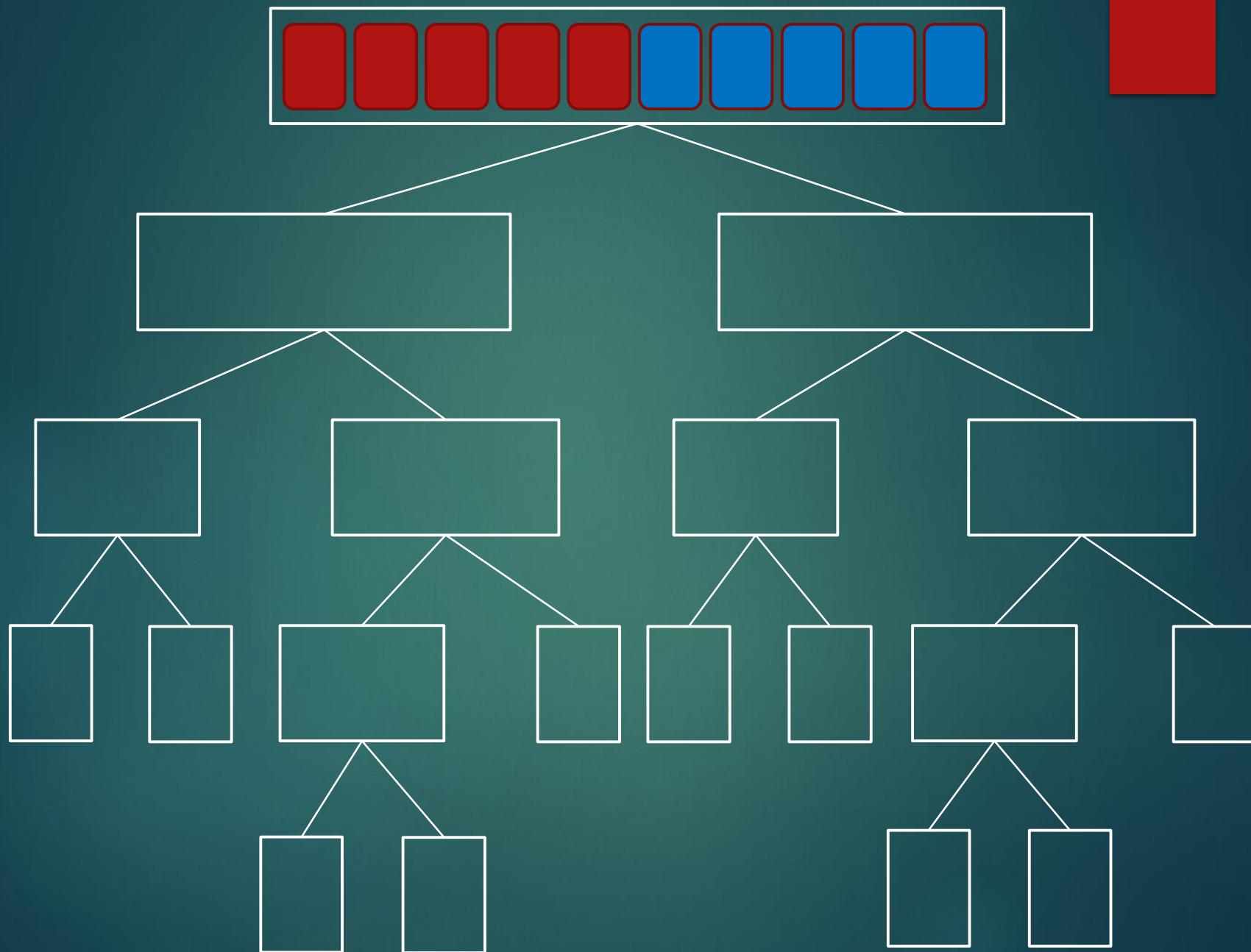
Merge Sort

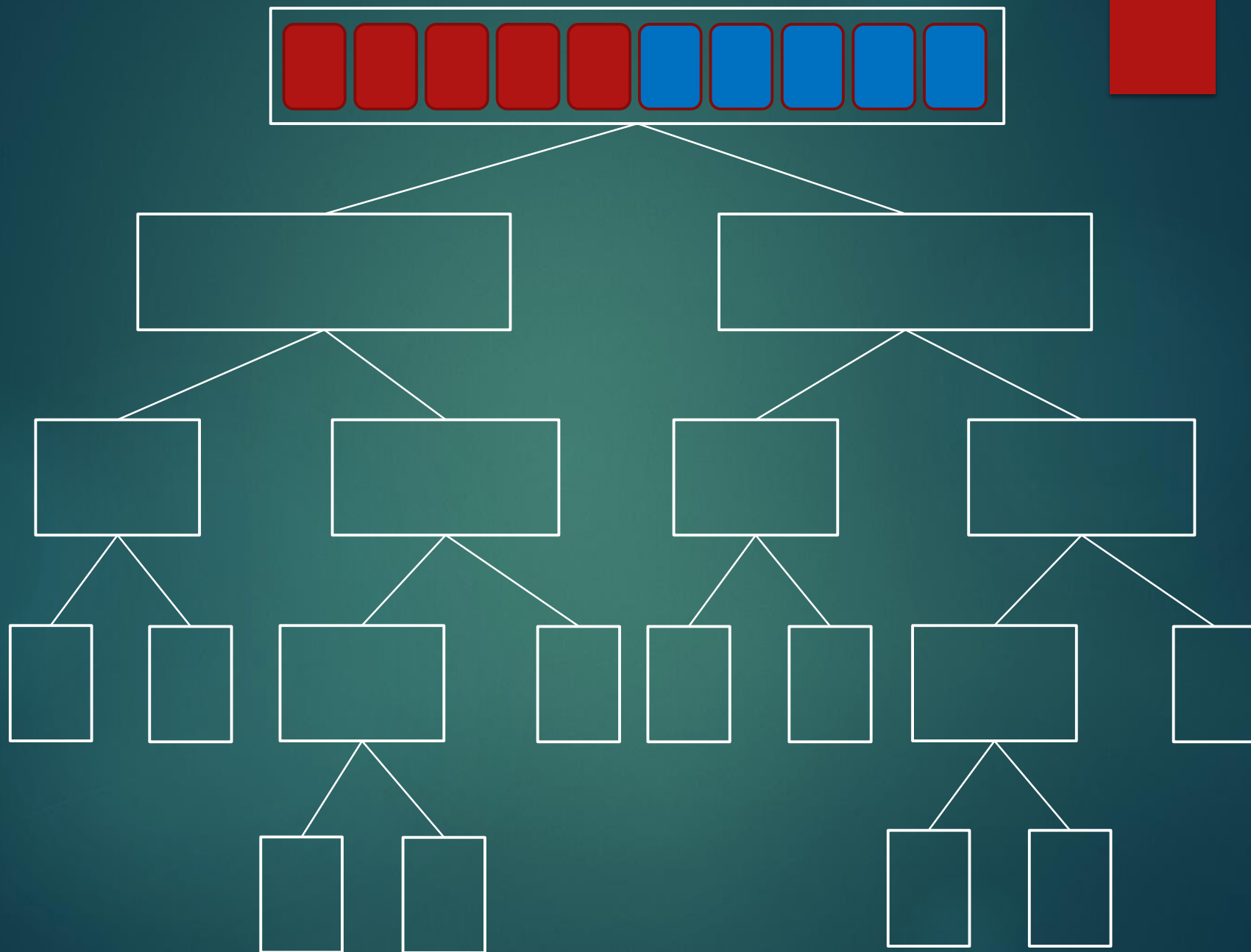
UNPLUGGED ACTIVITY

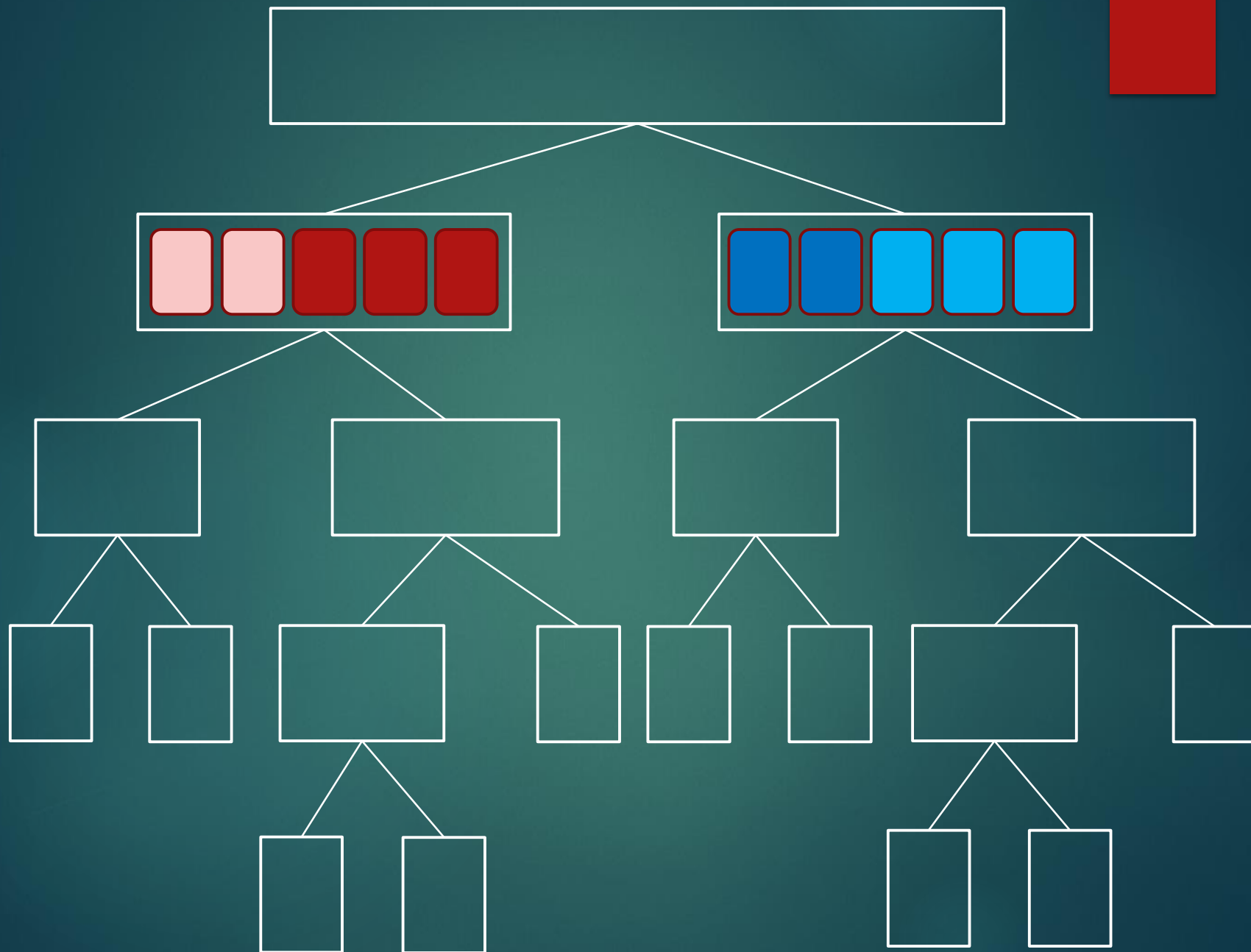
Unplugged Activity:

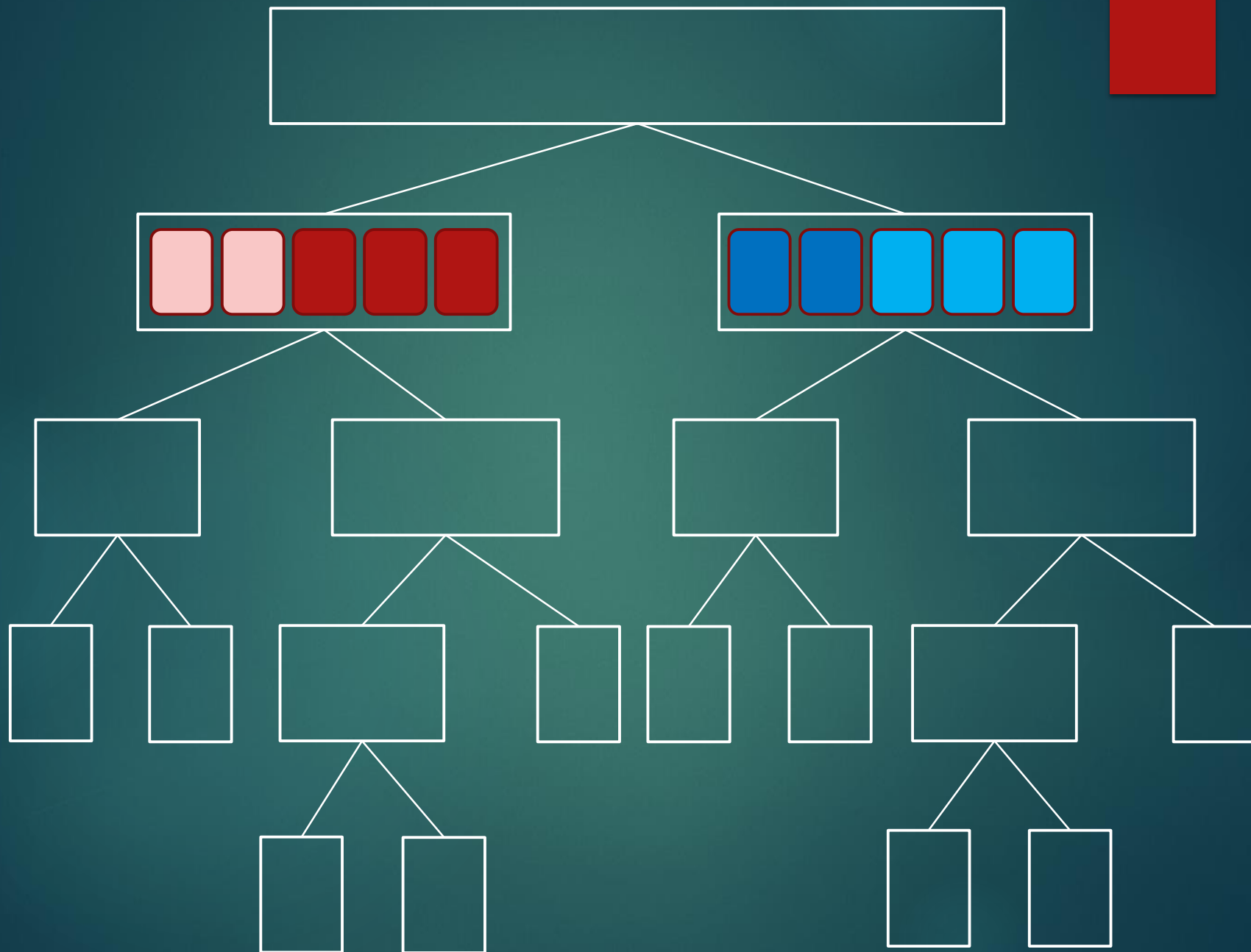
1. Start with a stack of 10 cards, all facing down on the table.
2. Divide the cards into 2 groups, then sub-divide the 2 groups again into further 2 groups ... Until every group contain only 1 card.
3. Compare the values of cards between the 2 'neighbouring' groups, and merge them back together. Repeat this until only 1 group left.
4. The cards in the group is sorted.

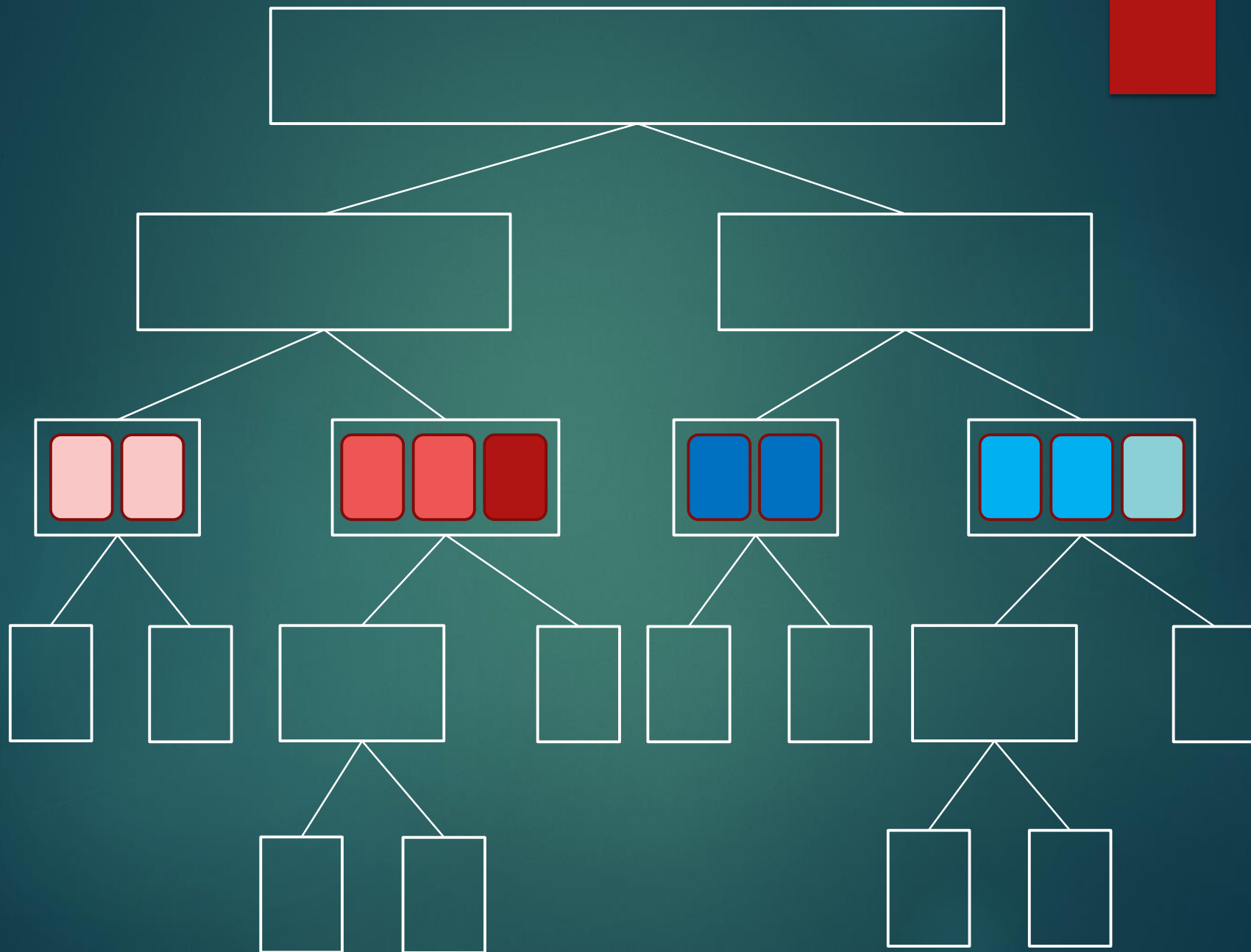


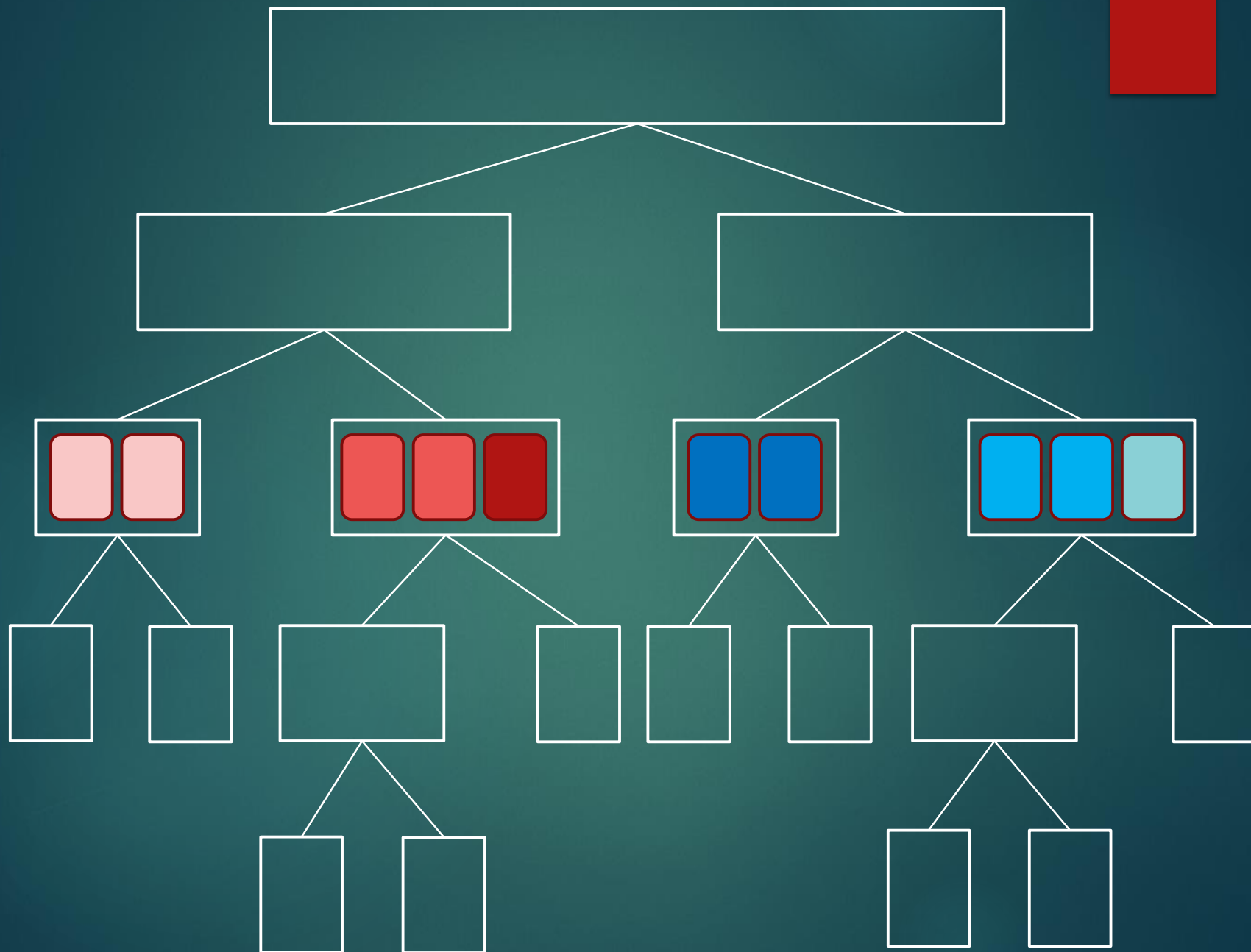


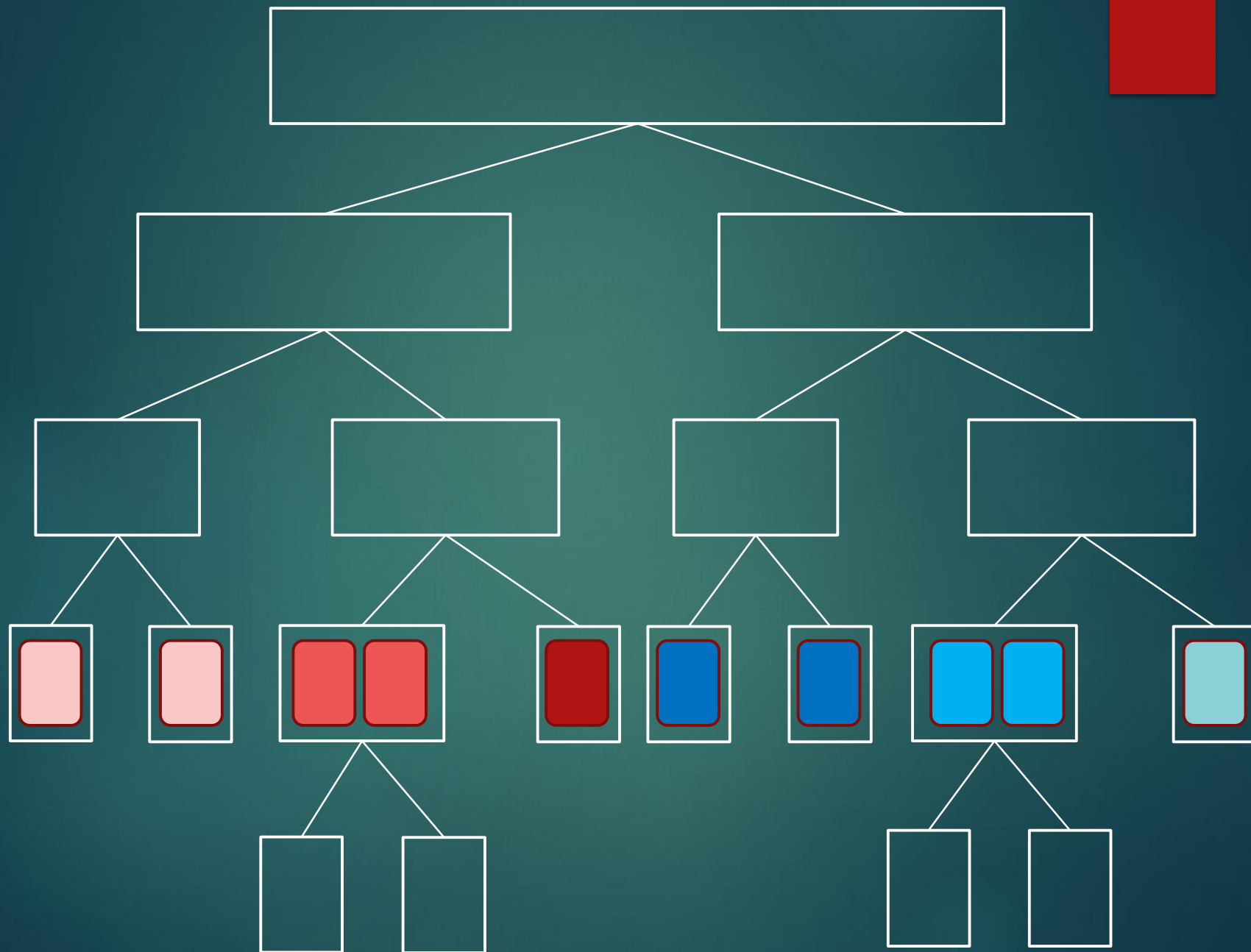


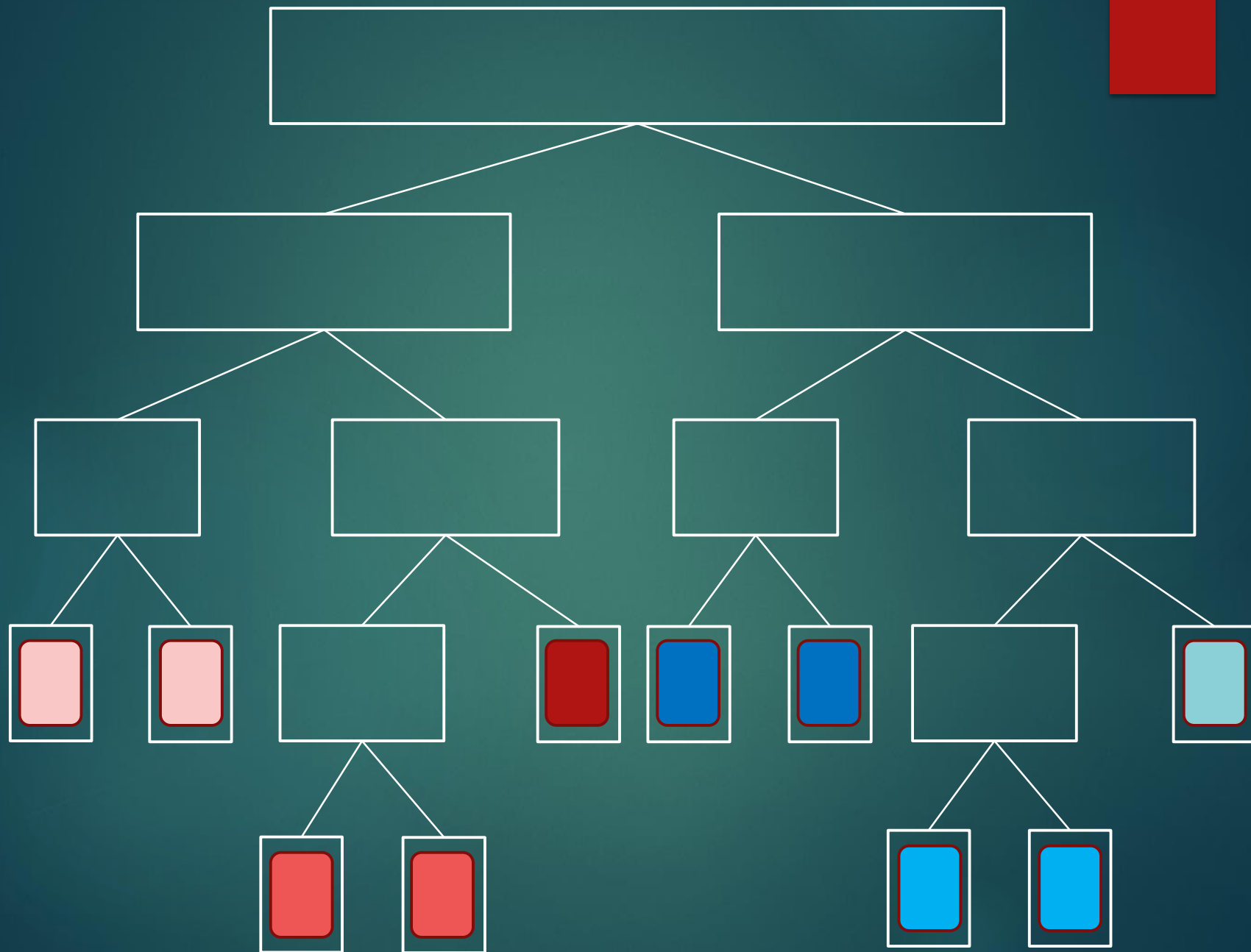


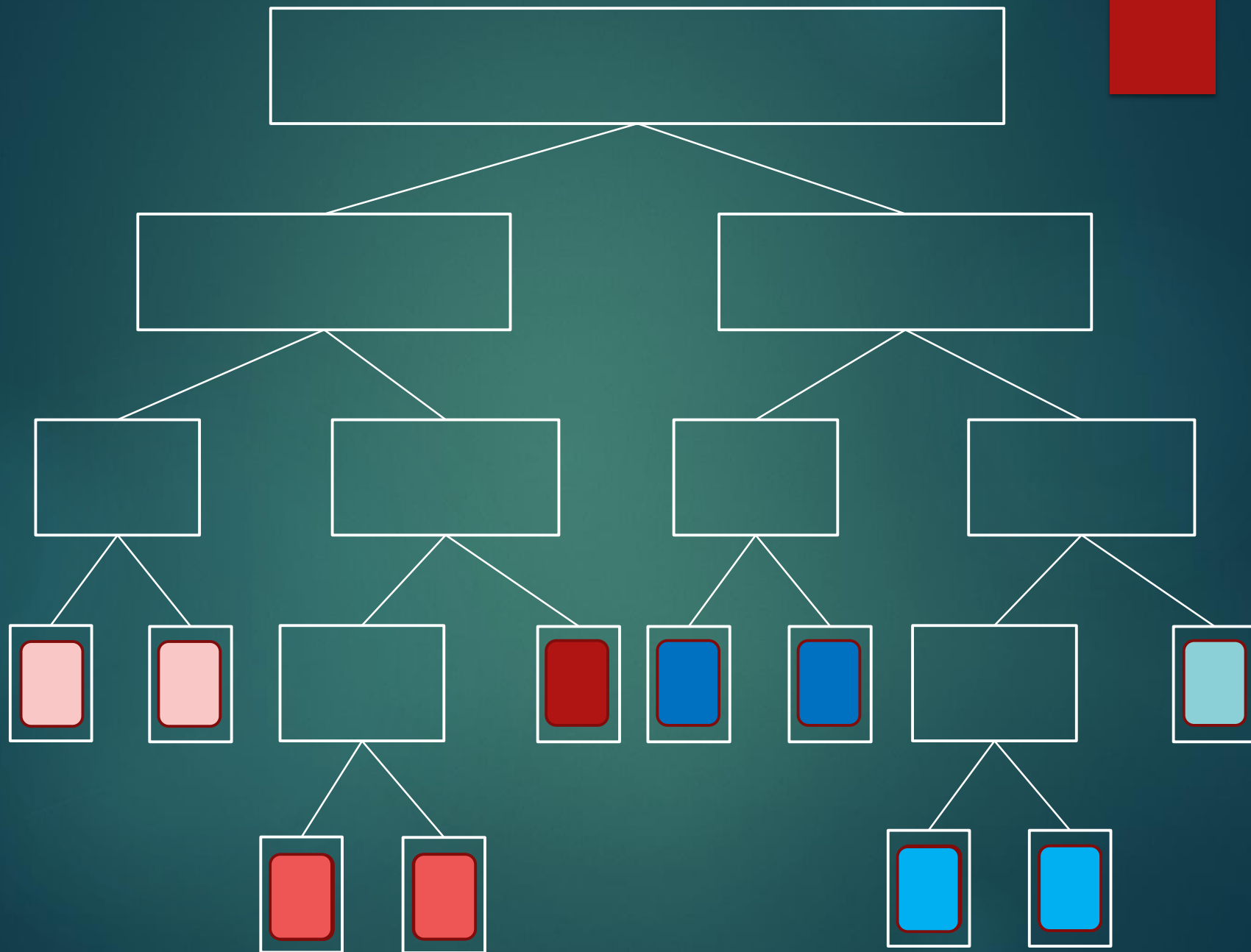


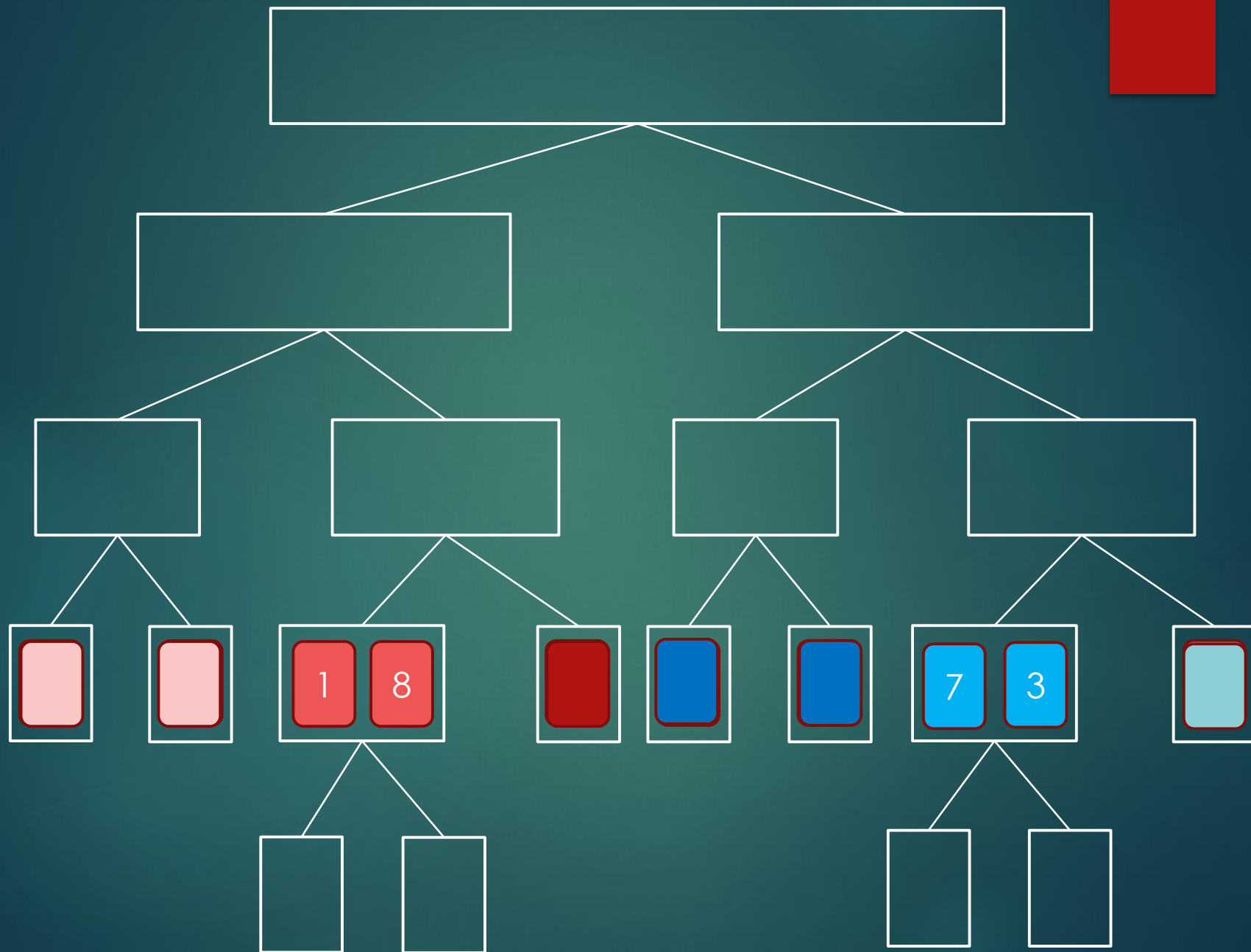


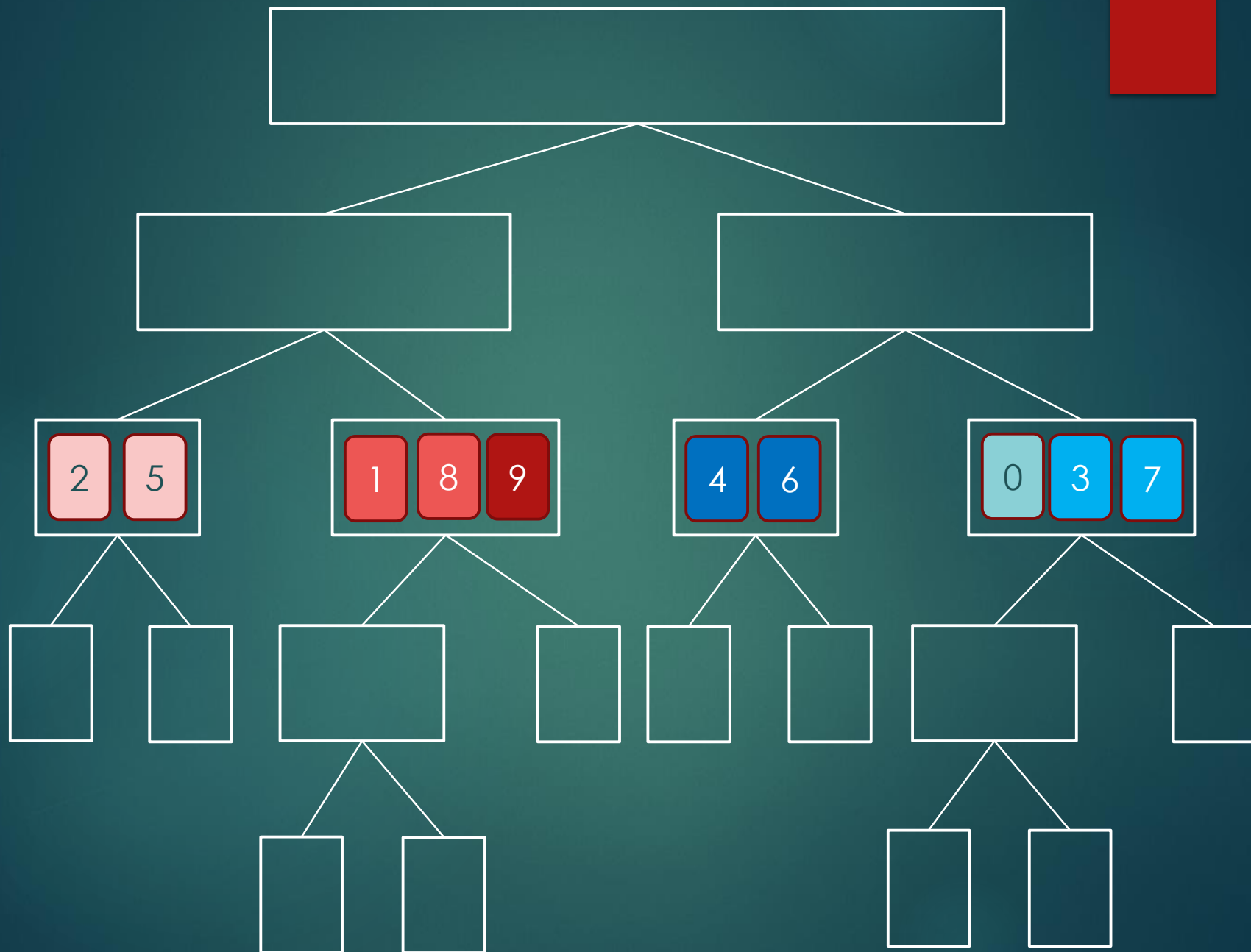


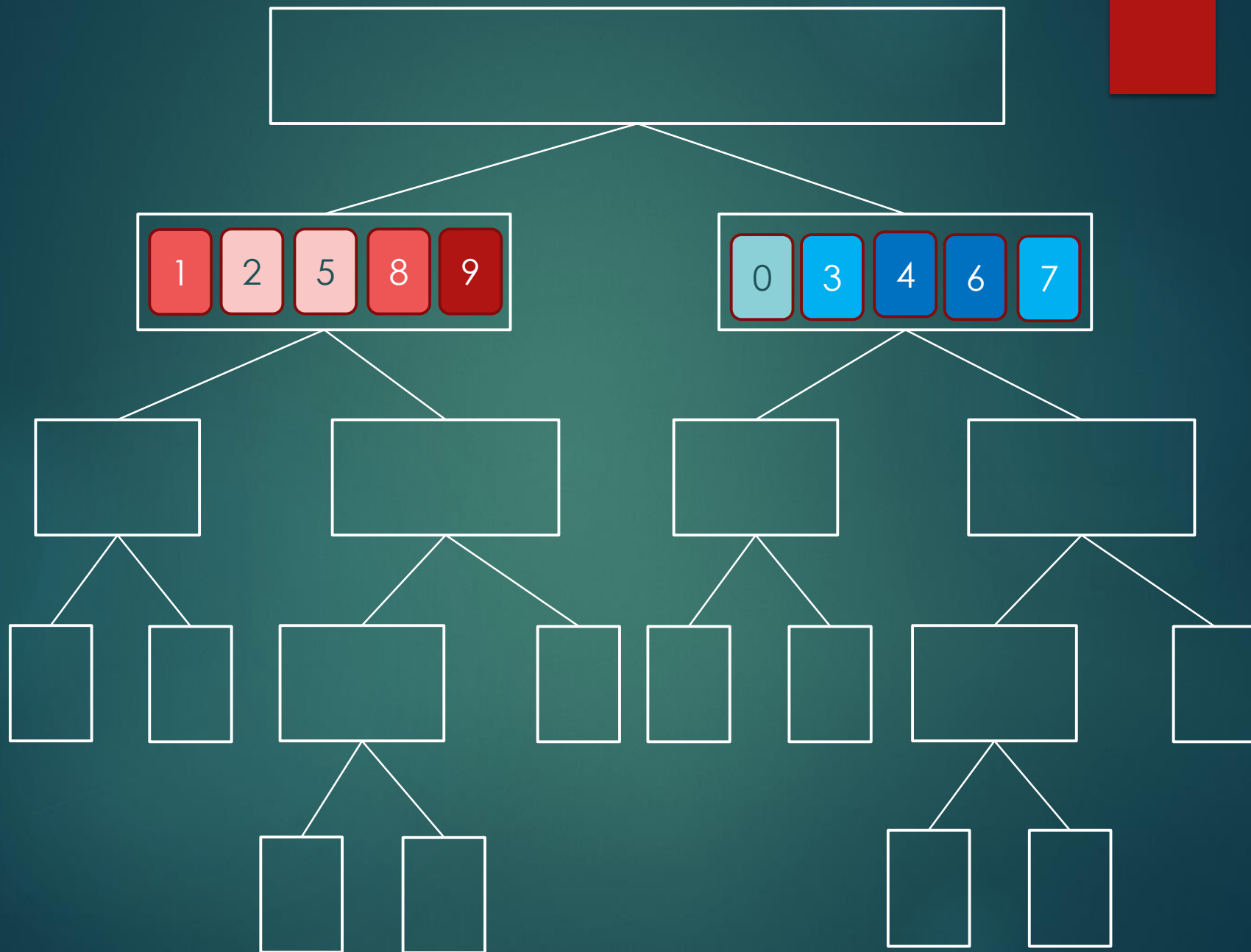


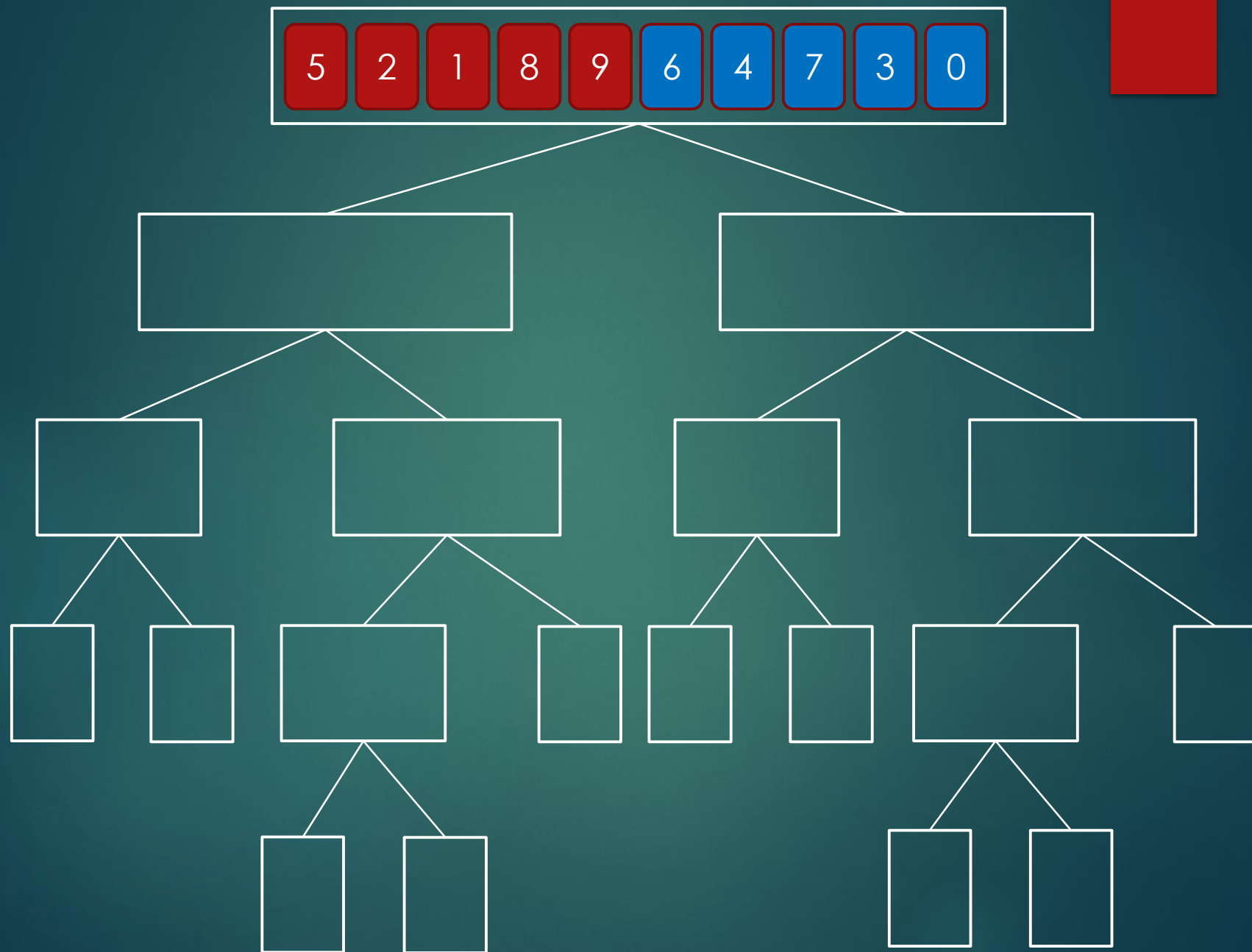














LT12d

Merge Sort

ALGORITHM

Description :

The **Merge Sort** algorithm will first divide the list into half over and over again until each list contains only one element.

Then it will merge two lists together into a sorted list repeatedly until all the elements are combined together in one single sorted list.

Basic Skills :

1. Split a list recursively
2. Merge two lists together

Illustration :

- ▶ Step 1: Start by dividing the array into two, again and again until every array contain only one element.

(5, 2, 1, 8, 9)

-> ([5, 2], [1, 8, 9])

-> (([5], [2]), ([1], [8, 9]))

-> (([5], [2]), ([1], ([8], [9])))

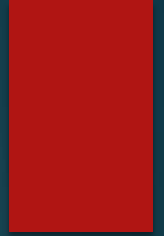
- ▶ Step 2: Merge them back in sorted order:

-> (([2, 5]), ([1], [8, 9]))

-> ([2, 5], [1, 8, 9])

-> ([1, 2, 5, 8, 9])

Merge Sort : Order of Growth



Best Case : $O(n \log n)$

Worst Case : $O(n \log n)$

Average Case : $O(n \log n)$